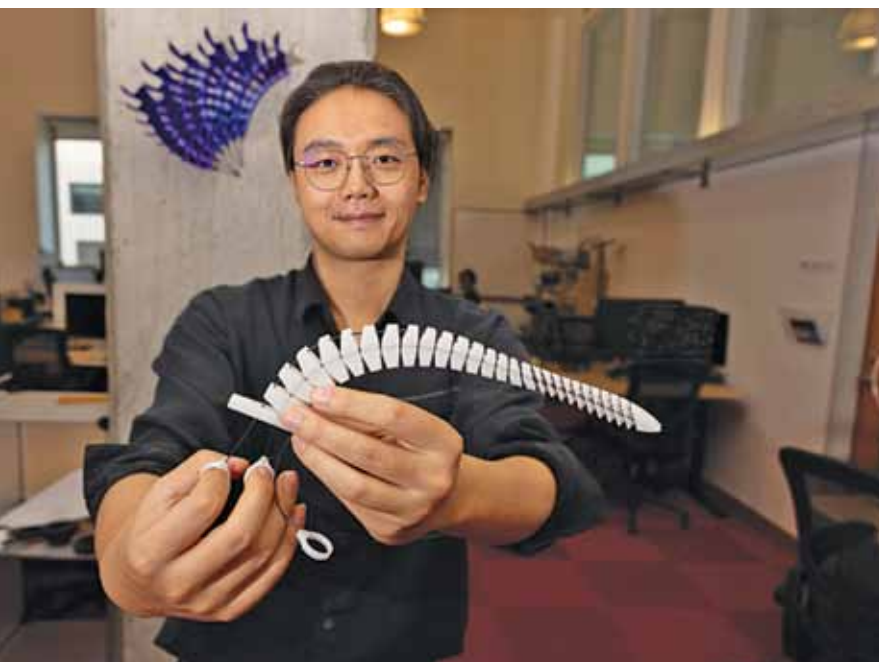


3D Printing Cable-Driven Robots With Precision



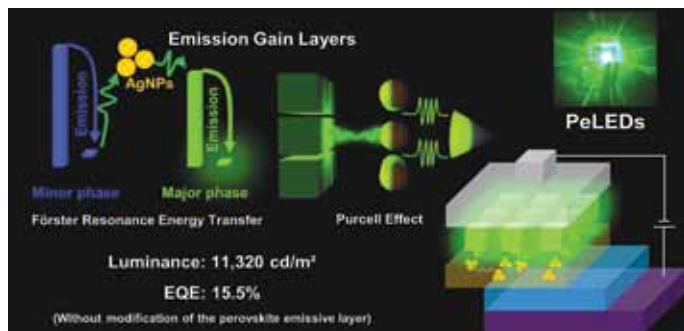
MIT's CSAIL (United States) has introduced Xstrings, a 3D printing method that embeds cables directly into printed structures, simplifying the creation of cable-driven devices. Using a multi-material printer operating at 260°C and 10-20mm/s, Xstrings produces flexible interiors and rigid exteriors in one step, cutting assembly time by 40%. It enables precise motion types like bending or coiling, demonstrated in creations like a lizard robot, tentacle, and robotic claw. Tested for over 60,000 cycles, it shows high durability. It could revolutionise robotics, wearable tech, and even space-based manufacturing.

The 'Xstrings' method can produce a range of colourful and unique objects, like a white tentacle that curls around items and a purple wall sculpture that can open and close (Credit: Mike Grimmitt/MIT CSAIL)

Nanotech Boosts PeLED Efficiency

Researchers from the Institute of Polymer Science and Engineering at the National Taiwan University (Taiwan), and the Institute of Organic and Polymeric Materials at the National Taipei University of Technology (Taiwan) have boosted perovskite light-emitting diode (PeLED) efficiency without altering the emissive layer by introducing a polycaprolactone-silver nanoparticle (PCL@AgNPs-P) emission gain layer. Using microwave-assisted post-addition, the layer leverages Förster resonance energy transfer (FRET) and the Purcell effect to enhance energy transfer and spontaneous emission. PeLEDs with this layer achieved 11,320cd/m² luminance and 15.5% EQE. This method maintains perovskite purity, improves fabrication stability, and delivers narrow, stable green emission (94meV), offer-

ing scalable potential for advanced displays, laser optics, and quantum photonics.



The newly developed polycaprolactone@silver nanoparticle layer significantly enhances energy transfer and spontaneous emission in perovskite LEDs (Credit: National Taiwan University)